THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. A sound wave travels from air (340 m/s) into water (1500 m/s). A 500 Hz wave has wavelengths in air and water of: 1

(a) 0.68 m and 0.68 m **(b)** 3 m and 0.68 m
(c) 0.68 m and 3 m **(d)** 3 m and 3 m

2. A cell is placed in a solution and its mass increases by 20 per cent. The solution is: 1

(a) isotonic **(b)** hypotonic to the cell
(c) hypertonic **(d)** saturated salt

3. A ring of bark is removed from a branch. After some days, the part above the ring swells due to: 1

(a) minerals **(b)** gas
(c) accumulation of water **(d)** accumulation of food

4. A new organism is found to be multicellular, with cell walls of chitin, non-motile, and with thread-like hyphae. Its kingdom is: 1

(a) Plantae **(b)** Protista **(c)** Fungi **(d)** Animalia

5. A woman's menstrual cycle is 28 days and menstruation begins on 1 March. The likely day of ovulation is: 1

(a) 1 March **(b)** 14 March **(c)** 28 March **(d)** 7 March

6. A surface of albedo 0.5 receives 800 W/m squared. The absorbed radiation per square metre in 10 s is: 1

(a) 4000 J **(b)** 400 J **(c)** 800 J **(d)** 8000 J

7. A scientist says 'The model fits in 99 of 100 trials, so the theory is proven'. A better statement is: 1

(a) the theory is proven forever
(b) the theory is strongly supported but remains open to tests
(c) the theory is false
(d) the data are wrong

8. A mixture can be separated by sublimation if: 1

(a) only one component sublimes **(b)** both sublime at equal rates
(c) neither component melts **(d)** both melt

9. A student heats 10 g of CaCO3. Only 5.6 g of solid CaO is left. The mass of CO2 evolved is: 1

(a) 4.4 g **(b)** 5.6 g **(c)** 10 g **(d)** 15.6 g

10. An unknown element has isotopes 24 (79 per cent), 25 (10 per cent) and 26 (11 per cent). The average atomic mass is: 1

(a) 24.9 u **(b)** 25 u **(c)** 24 u **(d)** 24.3 u

11. A boat goes 12 km downstream in 1 hour and returns upstream in 2 hours. The average speed for the round trip is: 1

(a) 9 km/h **(b)** 6 km/h **(c)** 12 km/h **(d)** 8 km/h

12. A book is placed on a table. The reaction to the weight of the book is: 1

(a) the pull of the book on the Earth **(b)** the normal force by the table
(c) the push of the table **(d)** friction

13. A block of mass 5 kg is pulled up a smooth incline of length 10 m to a height of 5 m. The work done against gravity is (g = 10): 1

(a) 500 J **(b)** 50 J **(c)** 250 J **(d)** 25 J

14. If the temperature rises from 0 degrees Celsius to 40 degrees Celsius, the speed of sound in air increases from 331 m/s by about (0.6 m/s per degree Celsius): 1

(a) 40 m/s **(b)** 6 m/s **(c)** 240 m/s **(d)** 24 m/s

15. Which of the following could not be a function of the Golgi apparatus? 1

(a) formation of lysosomes **(b)** packaging proteins
(c) secretion **(d)** synthesis of ATP

16. The tissue that produces the enzymes and hormones in glands is: 1

(a) bone **(b)** glandular epithelium
(c) muscle **(d)** squamous epithelium

17. **Assertion (A):** The displacement of a body can be zero even though the distance travelled is not zero. 1

Reason (R): Displacement is the shortest straight-line distance from the initial position to the final position, in a given direction.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Passengers in a bus jerk forward when the bus stops suddenly. 1

Reason (R): The lower part of the body stops with the bus, while the upper part tends to remain in motion because of inertia.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** No work is done by a person who holds a heavy suitcase while standing still. 1
Reason (R): Work is done only when a force produces a displacement of the body in the direction of the force.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Sound cannot travel through a vacuum. 1
Reason (R): Sound is a mechanical wave and needs a material medium for its propagation.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. A body of mass 2 kg is acted on by a force that changes its speed from 3 m/s to 5 m/s. Find the work done by the force. 2

22. A boy standing 85 m in front of a hill claps and hears the echo after 0.5 s. Find the speed of sound. 2

23. Why do cells divide when they reach a certain size? Use the surface area to volume ratio for a cube of side 1 cm and 3 cm. 2

24. Explain why wood is made mainly of xylem and why it is strong. 2

25. Why do scientists keep revising the classification of organisms? Give two reasons. 2

26. A cell in the human body has 46 chromosomes. State the number of chromosomes in a muscle cell, sperm, egg and zygote. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. A cube of potato of mass 10 g is placed in 0.5 per cent, 1 per cent and 5 per cent sugar solutions. Predict the changes with the reasoning and mention the solution that is isotonic if the mass is unchanged in the 1 per cent solution. 3

28. A potato disc and a carrot disc are placed in iodine solution. Predict the observation and identify the tissue involved. 3

29. Construct a simple key (two choices at each step) to identify the following: pigeon, frog, shark, bat, snake. 3

30. A woman has a 30-day menstrual cycle with menstruation starting on day 1. Calculate the probable day of ovulation, and the fertile period (taking the egg life as 1 day and sperm life as 3 days). 3

31. A household uses 250 units per month. Using 0.8 kg CO_2 per unit, calculate the annual emission and the number of trees (each absorbing 22 kg per year) needed to offset it. 3

OR

Explain why a scientist should report measurement uncertainties along with results with an example. 3

32. A student prepared two solutions: 5 g sugar in 50 g water (A) and 8 g sugar in 100 g water (B). Compare the concentrations. 3

OR

Explain why a compound with ions carrying greater charge such as MgO has a higher melting point than NaCl. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** In a school laboratory, 10 g of calcium carbonate is heated strongly in a closed hard-glass tube fitted with a gas collector. It decomposes: $CaCO_3 \longrightarrow CaO + CO_2$. The solid residue (calcium oxide) weighs 5.6 g. 5

(i) State the law of conservation of mass.

(ii) Find the mass of carbon dioxide formed.

(iii) Who proposed this law?

(iv) Another student heats the same carbonate in an open dish and finds that the residue is lighter than 10 g. Does this violate the law? Explain.

34. **Case study.** An atom of an element X has 11 protons and 12 neutrons. Meena is asked to describe it using what she has learnt about atomic structure. 5

(i) Find the atomic number and the mass number of X.

(ii) Write its electronic configuration.

(iii) What is its valency?

(iv) What happens to the electronic configuration when the atom forms an ion of valency 1, and how many electrons does the ion have?

35. Describe in detail an experiment to show that the seed germination needs water, air and a suitable temperature, with the set-up of 4 test tubes, controls and conclusions. 5

36. A student models the Earth's temperature by the balance of incoming and outgoing energy. The Sun delivers 342 W/m squared on average, the albedo is 0.3. Calculate the absorbed energy, the outgoing energy needed at equilibrium and state what happens if the albedo decreases to 0.28 and the effect of more greenhouse gases. 5

OR

Explain how a chemist plans an experiment on the effect of temperature on the rate of dissolving of sugar, including fair-test conditions, data recording, graph and analysis of errors. 5

37. Explain, with examples of colloids, the stability of colloids, coagulation, electrophoresis and dialysis. 5

OR

Explain why equal masses of different elements do not contain equal numbers of atoms. Compare 12 g of carbon with 12 g of magnesium (C = 12, Mg = 24) in terms of the numbers of atoms. 5

38. The nuclei of two elements A and B have the same number of neutrons (isotones), A has 12 protons and 24 u mass. Find the number of neutrons, the mass number of B if it has 11 protons, and the electronic configuration of B. 5

OR

Explain the motion of a ball moving in a vertical circle of radius 1 m with speed 4 m/s, the period, the speed vs velocity, and the average velocity over half a period (pi = 3.14). 5

END OF QUESTION PAPER **Total: 80 Marks**

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